CHAPTER 03 · SUPPORTED EDITION

Reproduction & Heredity

Why do you look like your family? The answer is in tiny instructions passed from parents to children. This guided edition walks you through traits, genes, and how living things make more of themselves — step by step.

📗 **Georgia Standard S7L3**

WHAT YOU'LL LEARN

📒 **CHAPTER WORD BANK**

trait · heredity · DNA · gene · chromosome · allele · dominant · recessive · clone · variation

The 🔑 boxes explain each word.

3.1 Traits and Heredity

A **trait** is something you can see about a living thing. Eye color and hair type are traits. **Heredity** means traits pass from parents to their children.

trait = a feature you can observe (eye color, height).
heredity = passing traits from parents to children.

You got instructions from **each** parent. These instructions decide many of your traits. That is why kids often look like their parents.

✓ Quick Check 3.1

1. What is a trait? Give one example.
2. What does heredity mean?

3.2 Genes and Chromosomes

> Your traits are written in a code called **DNA**. DNA fits inside every cell, like a set of nesting dolls.

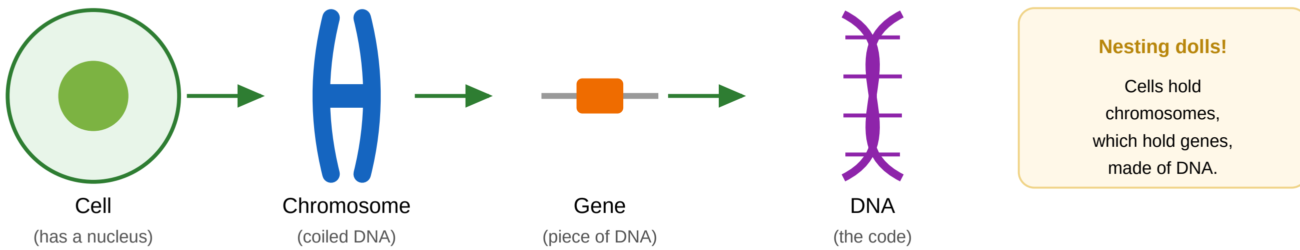

Figure 3.1 — From cell to DNA. Each part fits inside the part before it.

DNA = the code with all the instructions for life.
gene = a small piece of DNA that controls one trait.
chromosome = a long coil of DNA. Genes sit on it.

🔑 Remember the Order

A **cell** holds **chromosomes**. Chromosomes hold **genes**. Genes are made of **DNA**.

✓ Quick Check 3.2

1. What is a gene?
2. Humans get chromosomes from how many parents?

3.3 Strong and Weak Genes

Genes come in two versions called **alleles**. You get two — one from each parent. Some are **strong** (dominant). Some are **weak** (recessive).

dominant = the strong allele. Shown with a **CAPITAL** letter (T).
recessive = the weak allele. Shown with a **lowercase** letter (t).

Example — tongue rolling: **T** = can roll (strong). **t** = cannot roll (weak).

Genes	Result
TT	Can roll tongue
Tt	Can roll tongue (the strong T wins)
tt	Cannot roll tongue

🔑 **Big Rule**

A weak (recessive) trait only shows when you have **two** lowercase letters (tt).

3.4 Punnett Squares

A **Punnett square** is a tool that shows the chance of a child getting a trait. It has 4 boxes.

Parent 1: Tt

Parent 2: Tt

T	t
TT	Tt
Tt	tt

Cross: Tt × Tt

Fill each box with the letter on top and the letter on the side. Then count:

- **TT** = 1 box → can roll
- **Tt** = 2 boxes → can roll
- **tt** = 1 box → cannot roll

3 of 4 can roll. **1 of 4** cannot.

⭐ **Steps to Build One**

- Write one parent's letters on **top**.
- Write the other parent's letters on the **side**.
- Fill each box. Then count.

✓ **Quick Check 3.4**

1. In a Tt × Tt cross, how many of the 4 boxes can roll their tongue?

3.5 Asexual Reproduction

> **Asexual reproduction** needs only **one** parent. The baby is a **clone** — it has the same DNA as the parent.

clone = an exact copy. Same DNA as the parent.

Type	What Happens	Example
Binary fission	One cell splits into two	Bacteria
Budding	A bud grows off the parent	Hydra, yeast
Regeneration	A new body grows from a piece	Starfish

⭐ **Good and Bad**

Good: fast, no mate needed.
Bad: no variety — one disease could harm them all.

3.6 Sexual Reproduction

Sexual reproduction needs **two** parents. Each gives half its DNA. The baby is a mix — not the same as either parent. This makes **variation** (variety).

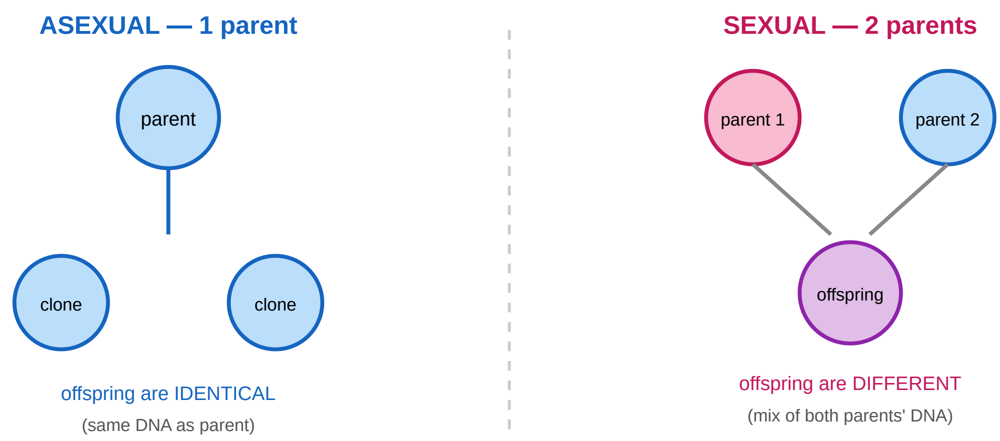

Figure 3.2 — One parent makes clones. Two parents make variety.

	Asexual	Sexual
Parents	One	Two
Baby's DNA	Same (clone)	A mix
Variety	None	Lots

✍️ Sentence Starter

Sexual reproduction makes variety because ________________________.

3.7 Selective Breeding

Selective breeding is when **people** choose which living things have babies, so a wanted trait shows up more often.

Started With	Became
Wild wolves	All dog breeds
Wild mustard plant	Broccoli, cabbage, kale
Tiny wild corn	Big sweet corn

! Key Difference

In selective breeding, **humans** pick the traits. In natural selection, **nature** picks.

✓ Quick Check 3.7

1. In selective breeding, who chooses which traits get passed on?

3.8 Review & Check

Great work! Here is everything in one place.

Big Idea	Remember This
Heredity	Traits pass from parents to children through genes.
Order	Cell → chromosome → gene → DNA.
Alleles	Strong (CAPITAL) beats weak (lowercase). Weak shows only as tt.
Punnett	A 4-box tool to show the chance of a trait.
Asexual vs sexual	One parent = clones. Two parents = variety.
Selective breeding	People choose the traits.

⭐ Standard Check — S7L3

You can now **(a)** explain how genes and chromosomes pass on traits, **(b)** show how asexual makes clones and sexual makes variety, and **(c)** describe selective breeding. 🎉